PS1 - IO

Tate Mason

```r
# Packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
# Data
iri <- read_tsv("~/SchoolWork/Y2S1/IO/PS1/IRI.csv", show_col_types = FALSE)
names(iri) <- gsub("^t", "", names(iri))  # Clean names
iri <- iri %>%
  mutate(
    market_id = interaction(store_id, week_id, drop = TRUE)
  )
```

1.1

```r
# Distinct counts per market
brands_per_market <- iri %>%
  distinct(market_id, brand) %>%
  count(market_id, name = "n_brands")
```

```r
mfrs_per_market <- iri %>%
  distinct(market_id, parent) %>%
  count(market_id, name = "n_mfrs")

# Total market sales per market
total_sales_market <- iri %>%
  group_by(market_id) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Brand-level sales (market x brand)
brand_sales_market <- iri %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Per-market price & characteristics (means within market)
market_chars <- iri %>%
  group_by(market_id) %>%
  summarise(
    price_mean     = mean(price, na.rm = TRUE),
    fiber_mean     = mean(fiber, na.rm = TRUE),
    sugars_mean    = mean(sugars, na.rm = TRUE),
    flavored_mean  = mean(as.numeric(flavored), na.rm = TRUE),
    fortified_mean = mean(as.numeric(fortified), na.rm = TRUE),
    .groups = "drop"
  )

# Helper to summarise numeric vectors
summarise_vec <- function(x) {
  tibble(
    mean   = mean(x, na.rm = TRUE),
    sd     = sd(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
    max    = max(x, na.rm = TRUE)
  )
}

# Tables for 1.1
tbl_counts <- bind_cols(
  variable = c("n_brands", "n_mfrs"),
  bind_rows(
    summarise_vec(brands_per_market$n_brands),
    summarise_vec(mfrs_per_market$n_mfrs)
```

```r
  )
)

tbl_total_sales <- bind_cols(
  variable = "total_sales (per market)",
  summarise_vec(total_sales_market$total_sales)
)

tbl_brand_sales <- bind_cols(
  variable = "brand_sales (per market-brand)",
  summarise_vec(brand_sales_market$brand_sales)
)

tbl_price_chars <- tribble(
  ~variable,                 ~mean, ~sd, ~median, ~max,
  "price (market mean)",
    summarise_vec(market_chars$price_mean)$mean,
    summarise_vec(market_chars$price_mean)$sd,
    summarise_vec(market_chars$price_mean)$median,
    summarise_vec(market_chars$price_mean)$max,
  "fiber (market mean)",
    summarise_vec(market_chars$fiber_mean)$mean,
    summarise_vec(market_chars$fiber_mean)$sd,
    summarise_vec(market_chars$fiber_mean)$median,
    summarise_vec(market_chars$fiber_mean)$max,
  "sugars (market mean)",
    summarise_vec(market_chars$sugars_mean)$mean,
    summarise_vec(market_chars$sugars_mean)$sd,
    summarise_vec(market_chars$sugars_mean)$median,
    summarise_vec(market_chars$sugars_mean)$max,
  "flavored share (market mean)",
    summarise_vec(market_chars$flavored_mean)$mean,
    summarise_vec(market_chars$flavored_mean)$sd,
    summarise_vec(market_chars$flavored_mean)$median,
    summarise_vec(market_chars$flavored_mean)$max,
  "fortified share (market mean)",
    summarise_vec(market_chars$fortified_mean)$mean,
    summarise_vec(market_chars$fortified_mean)$sd,
    summarise_vec(market_chars$fortified_mean)$median,
    summarise_vec(market_chars$fortified_mean)$max
)
```

```r
summary_11 <- bind_rows(
  tbl_counts,
  tbl_total_sales,
  tbl_brand_sales,
  tbl_price_chars
)

summary_11
```

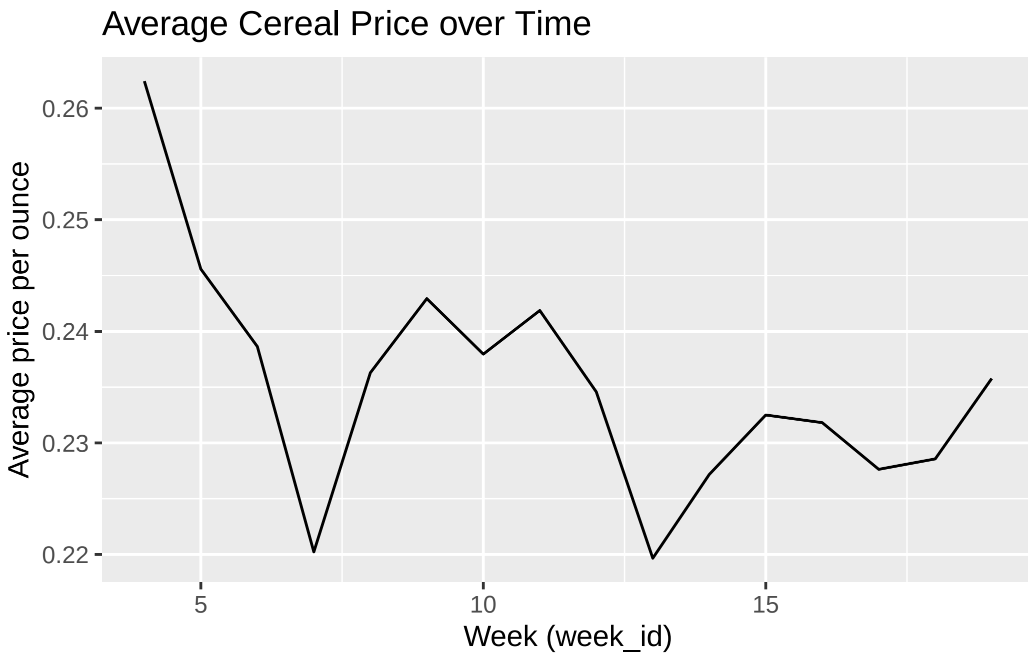
```
# A tibble: 9 x 5
  variable                              mean        sd    median       max
  <chr>                                <dbl>     <dbl>     <dbl>     <dbl>
1 n_brands                            34.9      4.82      37        39
2 n_mfrs                               3.95     0.259      4         4
3 total_sales (per market)        15555.    12358.     12557.    84567.
4 brand_sales (per market-brand)    446.      726.       238.    26470.
5 price (market mean)                  0.235    0.0265     0.233     0.325
6 fiber (market mean)                  1.30     0.0874     1.31      1.67
7 sugars (market mean)                 8.31     0.205      8.33     10.2
8 flavored share (market mean)         0.640    0.0265     0.641     0.857
9 fortified share (market mean)        0.485    0.0341     0.487     0.667
```

```r
# If you want a file:
# write_csv(summary_11, "summary_11_descriptives.csv")
```

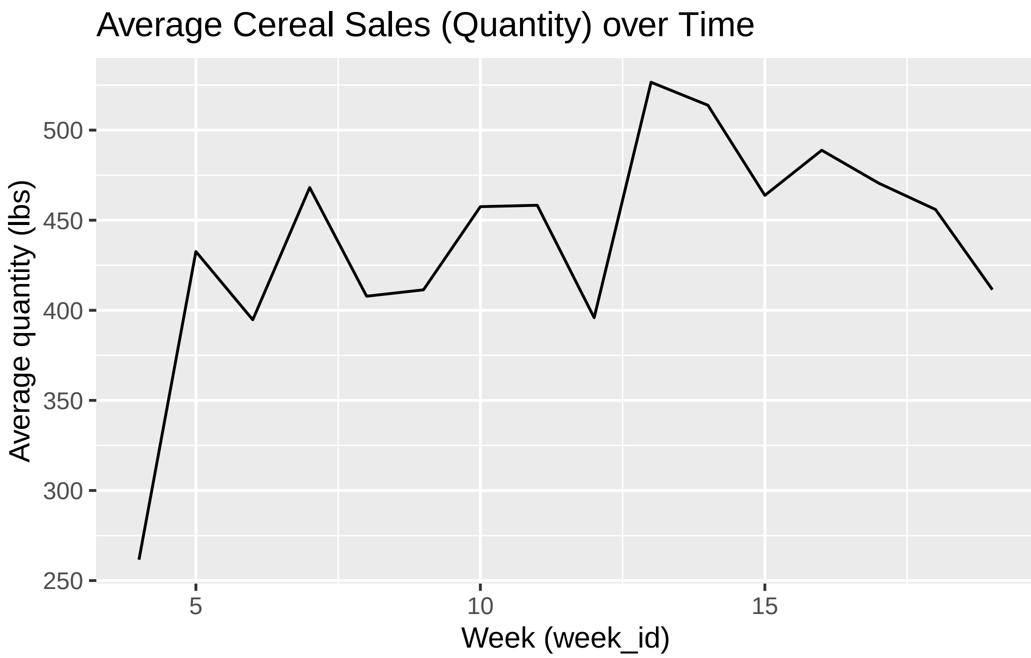
1.2

```r
ts_simple <- iri %>%
  group_by(week_id) %>%
  summarise(
    avg_price = mean(price, na.rm = TRUE),
    avg_qty   = mean(quantity, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(week_id)

p_price <- ggplot(ts_simple, aes(x = week_id, y = avg_price)) +
  geom_line() +
  labs(title = "Average Cereal Price over Time",
```

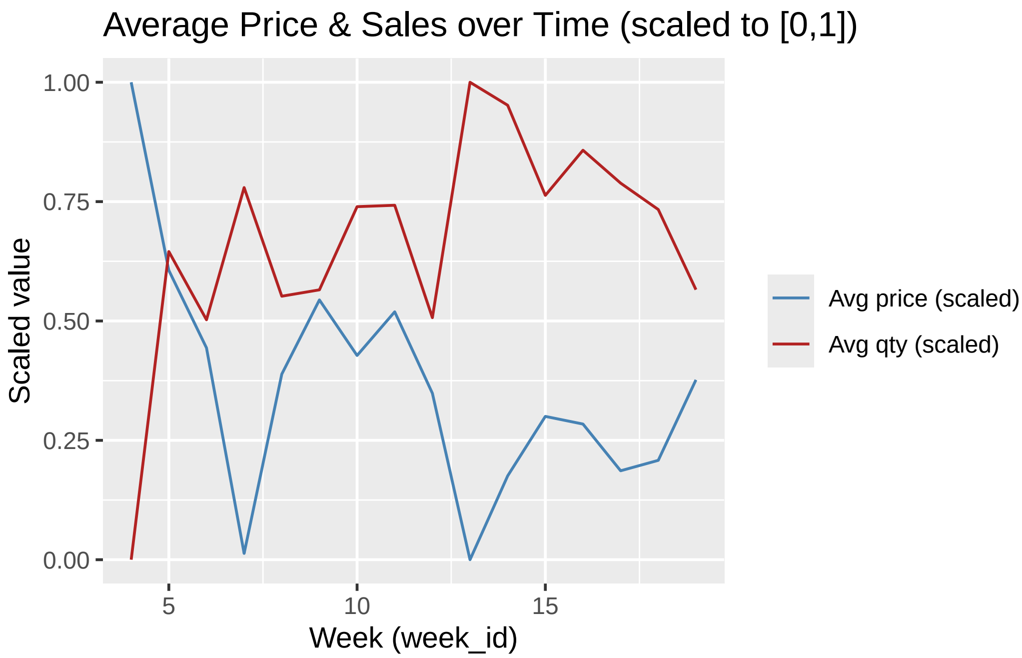
```r
       x = "Week (week_id)", y = "Average price per ounce")

p_qty <- ggplot(ts_simple, aes(x = week_id, y = avg_qty)) +
  geom_line() +
  labs(title = "Average Cereal Sales (Quantity) over Time",
       x = "Week (week_id)", y = "Average quantity (lbs)")

p_price
```




```r
p_qty
```




```
# Save figures
ggsave("avg_price.png", p_price, width = 10, height = 6, dpi = 300)
ggsave("avg_sales.png", p_qty, width = 10, height = 6, dpi = 300)

# Dual-axis style (scaled)
ts_long_scaled <- ts_simple %>%
  mutate(
    price_scaled = (avg_price - min(avg_price, na.rm = TRUE)) /
                   (max(avg_price, na.rm = TRUE) - min(avg_price, na.rm = TRUE)),
    qty_scaled   = (avg_qty - min(avg_qty, na.rm = TRUE)) /
                   (max(avg_qty, na.rm = TRUE) - min(avg_qty, na.rm = TRUE))
  ) %>%
  select(week_id, price_scaled, qty_scaled) %>%
  pivot_longer(-week_id, names_to = "series", values_to = "value")

p_dual <- ggplot(ts_long_scaled, aes(x = week_id, y = value, color = series)) +
  geom_line() +
  scale_color_manual(values = c("price_scaled" = "steelblue", "qty_scaled" = "firebrick"),
                     labels = c("Avg price (scaled)", "Avg qty (scaled)")) +
  labs(title = "Average Price & Sales over Time (scaled to [0,1])",
       x = "Week (week_id)", y = "Scaled value", color = NULL)

p_dual
```

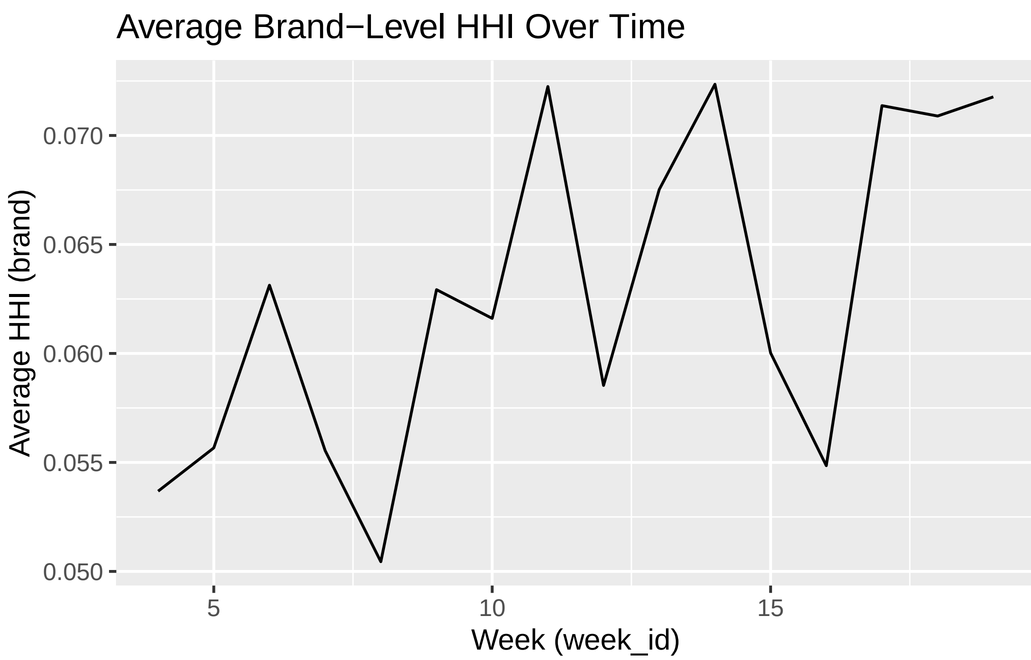



```
ggsave("avg_price_sales_dual.png", p_dual, width = 10, height = 6, dpi = 300)
```

1.3

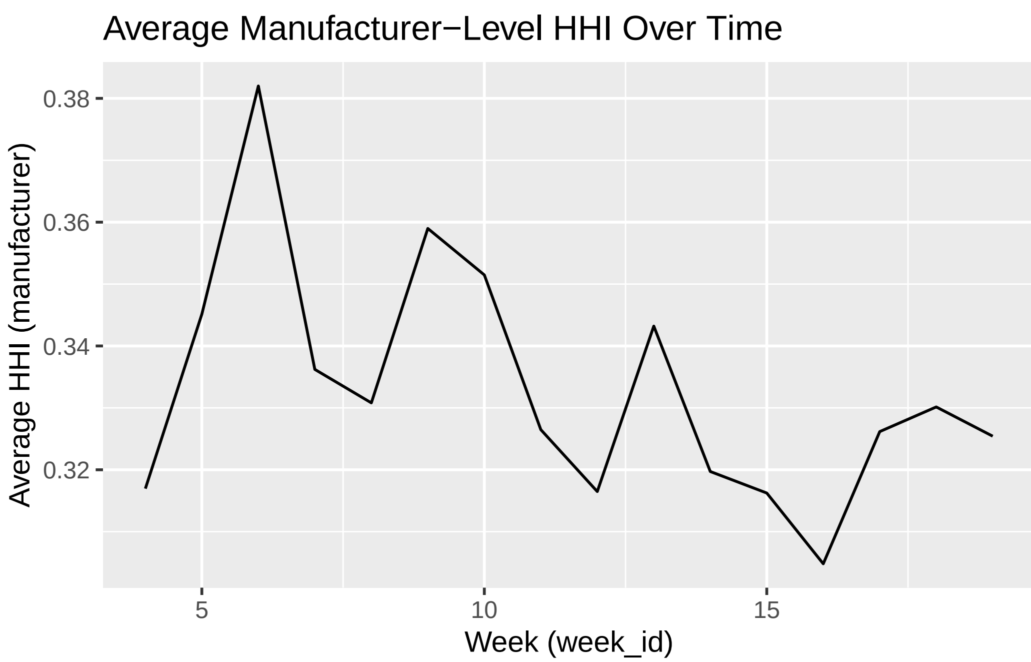
```
# Brand HHI per market
brand_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%
  mutate(share_brand = if_else(total_mkt_sales > 0, brand_sales / total_mkt_sales, 0))

hhi_brand_market <- brand_shares %>%
  group_by(market_id) %>%
  summarise(HHI_brand = sum(share_brand^2, na.rm = TRUE), .groups = "drop")

hhi_brand_week <- iri %>%
  distinct(market_id, week_id) %>%
```

```
  inner_join(hhi_brand_market, by = "market_id") %>%
  group_by(week_id) %>%
  summarise(avg_HHI_brand = mean(HHI_brand, na.rm = TRUE), .groups = "drop") %>%
  arrange(week_id)

p_hhi_brand <- ggplot(hhi_brand_week, aes(x = week_id, y = avg_HHI_brand)) +
  geom_line() +
  labs(title = "Average Brand-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (brand)")

p_hhi_brand
```



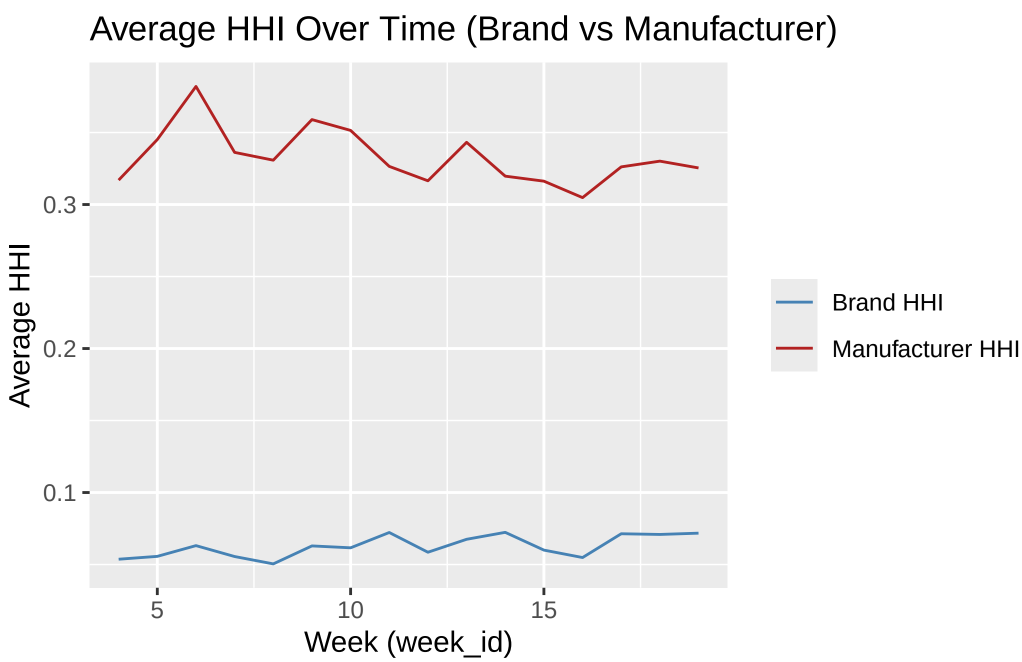

```
ggsave("HHI_brand.png", p_hhi_brand, width = 10, height = 6, dpi = 300)

# Manufacturer (parent) HHI per market
parent_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, parent) %>%
  summarise(parent_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%
```

```
  mutate(share_parent = if_else(total_mkt_sales > 0, parent_sales / total_mkt_sales, 0))

hhi_parent_market <- parent_shares %>%
  group_by(market_id) %>%
  summarise(HHI_parent = sum(share_parent^2, na.rm = TRUE), .groups = "drop")

hhi_parent_week <- iri %>%
  distinct(market_id, week_id) %>%
  inner_join(hhi_parent_market, by = "market_id") %>%
  group_by(week_id) %>%
  summarise(avg_HHI_parent = mean(HHI_parent, na.rm = TRUE), .groups = "drop") %>%
  arrange(week_id)

p_hhi_parent <- ggplot(hhi_parent_week, aes(x = week_id, y = avg_HHI_parent)) +
  geom_line() +
  labs(title = "Average Manufacturer-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (manufacturer)")

p_hhi_parent
```




```
ggsave("HHI_parent.png", p_hhi_parent, width = 10, height = 6, dpi = 300)

# Combined comparison
```

```r
hhi_both <- hhi_brand_week %>% inner_join(hhi_parent_week, by = "week_id")

p_hhi_both <- ggplot(hhi_both, aes(x = week_id)) +
  geom_line(aes(y = avg_HHI_brand, color = "Brand HHI")) +
  geom_line(aes(y = avg_HHI_parent, color = "Manufacturer HHI")) +
  scale_color_manual(values = c("Brand HHI" = "steelblue", "Manufacturer HHI" = "firebrick")
  labs(title = "Average HHI Over Time (Brand vs Manufacturer)",
       x = "Week (week_id)", y = "Average HHI", color = NULL)

p_hhi_both
```




```r
ggsave("HHI_brand_vs_parent.png", p_hhi_both, width = 10, height = 6, dpi = 300)

# Optional numeric outputs
# write_csv(hhi_brand_week,  "hhi_brand_by_week.csv")
# write_csv(hhi_parent_week, "hhi_parent_by_week.csv")
# write_csv(hhi_both,        "hhi_brand_parent_by_week.csv")
```